

Mutually Exclusive Events

- When two events have _____, we refer to them as mutually exclusive events.
- $P(A \text{ and } B) = \underline{\hspace{2cm}}$
- $P(A \text{ or } B) = \underline{\hspace{2cm}}$

Example 1) Let's revisit our "rolling two dice" game, where the italicized numbers represent the sum of the two dice rolled.

Dice	1	2	3	4	5	6
1	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>
2	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
3	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>
4	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
5	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>	<i>11</i>
6	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>	<i>11</i>	<i>12</i>

A = Rolling a sum of 5

B = Rolling two even numbers

Are these events mutually exclusive?

$P(A) = \underline{\hspace{2cm}}$ and $P(B) = \underline{\hspace{2cm}}$

$P(A \text{ or } B) = \underline{\hspace{2cm}}$

Example 2) Identify if the following events are mutually exclusive. If they are not, give an example of where they overlap.

a) Rolling a sum of 4 and rolling doubles.

b) Rolling an odd sum and rolling a 3.

c) Rolling a 4 and rolling a sum of 12.

d) Rolling an odd number and rolling a sum of 10.

How do we find the probabilities of non-mutually exclusive events occurring together?

Example 3) Find the probability of rolling doubles or a sum of 6.

A = Rolling doubles

B = Rolling a sum of 6

Are these events mutually exclusive?

$P(A) = \underline{\hspace{2cm}}$ and $P(B) = \underline{\hspace{2cm}}$

$P(A \text{ and } B) =$

$P(A \text{ or } B) =$

Dice	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

When we have events that “overlap”, adding their individual probabilities counts the “overlap” twice, so we have to subtract out one of the overlaps so it only gets counted once! This is called the addition rule

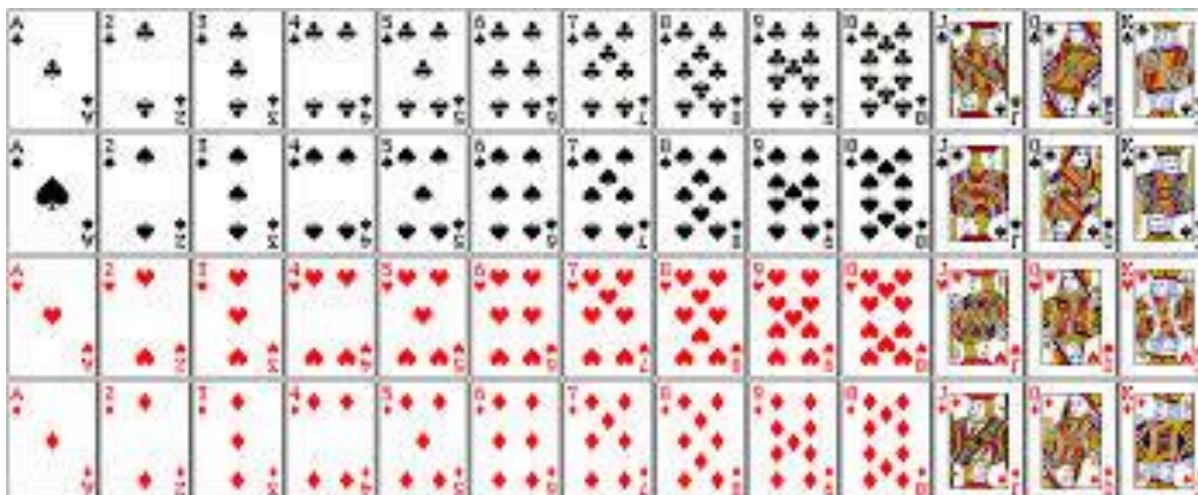
General Addition Rule

If A and B are any two events resulting from some chance process, then

Notice how if A and B are mutually exclusive, $P(A \text{ and } B) = \underline{\hspace{2cm}}$ and $P(A \text{ or } B) = \underline{\hspace{4cm}}$

How do we know the value of $P(A \text{ and } B)$? For now, we will use common sense and drawing out the sample space, but the *general multiplication rule* will allow us to find this mathematically later.

Example 4) A card is selected at random from a deck of 52 cards. Determine if the following events (on the next page) are mutually exclusive and then find the probability of the events.



(a) $P(\text{red or Queen})$

(b) $P(\text{Ace or King})$

(c) $P(\text{Queen or even number})$

(d) $P(\text{black or odd number})$

Example 5) You were interested in how many students at your school regularly eat breakfast. You conducted a survey, asking, “Do you eat breakfast on a regular basis?” Your random sample consisted of 600 students at the school and the results are shown in the table below.

	Male	Female	Total
Eats Breakfast Regularly	190	110	300
Doesn't Eat Breakfast Regularly	132	168	300
Total	322	278	600

If we select a student from this sample at random, what is the probability that the student is

(a) A female?

(b) Someone who eats breakfast regularly?

(c) A female and eats breakfast regularly?

(d) A female or eats breakfast regularly?